

Artificial F1: Full Computational Model

Selection Hardness, Non-Scalarizability, and Phase Transition in Bounded Evaluative Architectures

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May 2026

Abstract

We introduce a formal framework for bounded evaluative architectures in which signal admission precedes task specification. The central object is a gating operator $F_1: S \rightarrow S'$ that constitutes a selection-conditioned subspace $S' \subset S$ prior to any encoding, learning, or reward optimisation. We prove three main results. First, valence-conditioned evaluation over S' is non-scalarizable (Theorem 2A-1): the multi-dimensional preference structure induced by selection-conditioned access violates the totality condition required for scalar representation, and selection-conditioning generates incommensurability — a strictly stronger failure — across distinct selection-conditioned submanifolds. Second, the evaluative state $F_2(t)$ satisfies a stochastic differential equation whose stationary distribution and stability properties depend on the selection hardness $H_s(G)$ and noise variance σ^2 (Theorems 2A-2, 2A-3). Third, the action-selection entropy exhibits a phase transition at the gating threshold: a discontinuity that cannot arise in scalar-reward systems with smooth policies, and that is architecture-invariant below the collapse threshold γ^* while architecture-dependent above it. We introduce Selection Hardness $H_s(G)$ as a measurable quantity connecting these theoretical results to empirical predictions developed in the companion paper (Paper 2B). This paper makes no empirical claims; all experimental validation is deferred to Paper 2B.

Keywords: selection-conditioned representation, non-scalarizability, selection hardness, phase transition, gating architecture, bounded evaluation, RLHF, dynamical systems, representation constitution

2A-1. Introduction and Positioning

Reinforcement learning and preference optimisation share a foundational assumption that is rarely made explicit: that evaluation can be represented as a scalar function over states and actions. Reward functions map state-action pairs to real numbers; RLHF reduces human preference to a scalar comparison signal; multi-objective methods aggregate competing criteria into a single weighted sum. The scalar assumption is not merely a computational convenience. It is a claim about the structure of evaluation — that the preference ordering induced by any evaluation system can be represented by a single real-valued function without structural loss.

This paper proves that this claim fails for a specific and important class of systems: those in which evaluation operates on a selection-conditioned representation rather than on a fixed input space. The failure is not behavioural — it cannot be detected by observing input-output relationships. It is

structural: the mathematical conditions required for scalar representability are violated by the geometry of selection-conditioned evaluation, independently of any particular learning algorithm or reward specification. We call this failure Non-Scalarizability of Valence-Conditioned Evaluation (Theorem 2A-1).

The argument proceeds in four steps. First, we formalise the distinction between evaluation as a function and evaluation as a state: the evaluative system $F_2(t) = E(F_1(t), \theta)$ is a dynamical object that evolves over time and depends on a selection-conditioned input, not a fixed mapping from states to scalars. Second, we introduce Selection Hardness $H_s(G)$ as a measurable quantity characterising the information-destruction rate of the gating operator. Third, we prove that when $H_s(G)$ exceeds a critical threshold, the action-selection entropy exhibits a phase transition that cannot arise in scalar-evaluation systems. Fourth, we show that collapse dynamics are architecture-invariant below the threshold while phase behaviour is architecture-dependent above it.

This paper makes no empirical claims and reports no experiments. It is a mathematical paper whose results are conditional on the formal system specified in Section 2A-3. Experimental validation is developed in full in Paper 2B.

The paper is structured as follows. Section 2A-2 positions the contribution relative to existing work. Section 2A-3 specifies the formal system. Section 2A-3.5 introduces Selection Hardness. Section 2A-4 states and proves Theorem 2A-1. Section 2A-5 develops the SDE formulation. Sections 2A-6 and 2A-7 establish the variational objective and fixed-point existence. Section 2A-8 proves the Phase Transition theorem. Section 2A-9 proves Dimensionality Reduction. Section 2A-10 discusses open problems.

2A-2. Novelty Positioning and Related Work

The contribution of this paper is distinct from existing work in seven specific respects, detailed in Sections 2A-2.1 through 2A-2.7.

2A-2.1 Acknowledged Conceptual Overlap

The present framework shares vocabulary and motivation with RLHF (Christiano et al., 2017; Ziegler et al., 2019; Bai et al., 2022), multi-objective RL (Liu et al., 2015; Roijers et al., 2013), predictive processing (Friston, 2010; Clark, 2016), and the information bottleneck (Tishby et al., 2000). None of these programmes addresses whether evaluation can be represented as a scalar function when it operates on a selection-conditioned subspace.

2A-2.2 Evaluation as State, Not Function

In standard RL and RLHF, reward is a function $r: X \times A \rightarrow \mathbb{R}$ defined over the full state space. In the present framework, the evaluative state $F_2(t) = E(F_1(t), \theta)$ depends on the history of selections and evolves dynamically. This distinction implies that evaluation cannot be represented as a function over the full state space without losing the selection-dependence that gives it its structure — which is precisely what Theorem 2A-1 proves.

2A-2.3 Selection-Conditioned Representation

Attention mechanisms (Vaswani et al., 2017) and sparse coding (Olshausen & Field, 1997) perform selective processing within a fixed representational manifold. The F1 gating operator produces $S' = F_1(S)$ with $\dim(S') \ll \dim(S)$ — a different manifold entirely. The difference is topological, not parametric: attention produces a differently-weighted version of the same manifold; F1 produces a different manifold. No reparameterisation of attention recovers this reduction.

2A-2.4 Non-Scalarizability as Structural Result

Multi-objective RL impossibility results (Rojers et al., 2013) concern behavioural optimisation. Theorem 2A-1 is a structural result concerning the mathematical form of evaluation, proved via Debreu's (1954) representation theorem. The proof establishes incommensurability — not merely incomparability — across distinct selection-conditioned submanifolds. This is the paper's primary novelty.

2A-2.5 Dynamical Systems Perspective

The evaluative state $F_2(t)$ evolves according to a stochastic differential equation whose stationary distribution depends on the selection geometry and noise structure. The phase transition result (Theorem 2A-3) is a consequence of Jacobian eigenvalue structure changes as $H_s(G)/\sigma^2$ crosses γ^* . This has no analogue in RLHF or multi-objective RL literature.

2A-2.6 Summary Comparison Table

Framework	Evaluation form	Input space	Scalar?	Phase trans.
Standard RL	$r: X \times A \rightarrow \mathbb{R}$	Fixed S	Yes	No
RLHF	Pairwise σ	Fixed S	Yes (implicit)	No
Multi-obj RL	$r_1, \dots, r_k \in \mathbb{R}$	Fixed S	By aggregation	No
Attention	Weighted sum	Fixed S (reweighted)	Yes	No
Present work	$F_2(t) \in \mathbb{R}^p$ (dynamical)	$S' = F_1(S) \subset S$	No (Thm 2A-1)	Yes (Thm 2A-3)

Table 2A-1. Framework comparison across evaluation form, input space, scalar representability, and phase transition behaviour.

2A-2.7 Representation Constitution vs. Representation Analysis

Pohl et al. (2024, 2026) distinguish four dimensions of neural representation: sensitivity, specificity, independence, sufficiency. Their framework analyses representations within a given manifold. The present framework addresses what is prior: the constitution of the representational manifold itself through selection-conditioned access. Constitution is not a fifth dimension of their framework; it is the upstream process that determines the space within which their four dimensions apply. This distinction grounds the topological non-equivalence claim of Section 2A-2.3 and connects the phase transition (Section 2A-8) to the sufficiency dimension of Pohl et al.: the collapse threshold is the point at which the constituted manifold becomes causally sufficient for the target behaviour.

2A-3. Formal System Definition

We define the minimal formal system required to state the main theorems. All results are conditional on this specification; extensions and relaxations are noted in Section 2A-10.

Signal Space and Encoding

Let S be a compact metric space of raw signals, equipped with a Borel probability measure μ_S representing the natural signal distribution. The encoder $\Phi: S \rightarrow \mathbb{R}^n$ is a Lipschitz map with constant L_Φ , producing an n -dimensional latent representation for each input. We write $F_1(x) = \Phi(x) \in \mathbb{R}^n$ for the encoded signal. We assume signal decomposability: for any $x \in S$, the components $(F_1(x))_i$ are statistically exchangeable conditional on the signal class.

The gating operator $G: \mathbb{R}^n \rightarrow \{0,1\}^n$ is defined by a sparsity parameter $k \leq n$. For any input $z \in \mathbb{R}^n$, $G(z)$ selects the k components of z with largest absolute value, setting all others to zero. We denote the selection-conditioned subspace as $S' = \{G(F_1(x)) : x \in S\} \subset \mathbb{R}^n$, with $\dim(S') \leq k$. Functional separation is assumed: G depends on the magnitude of $F_1(x)$ and not on the evaluation state $F_2(t)$, ensuring the gate is upstream of evaluation.

Valence Tensor and Evaluative State

The valence function $v: S' \rightarrow \mathbb{R}^m$ with $m \geq 2$ assigns a multi-dimensional valence vector to each gated representation. The integer m is the valence dimensionality; $m \geq 2$ is the minimal condition for the non-scalarizability result. Typical instantiations include: $m = 2$ (arousal and valence), $m = 3$ (adding urgency), or higher-dimensional affect decompositions. We make no commitment to a specific m ; the theorem holds for all $m \geq 2$.

The evaluative state $F_2(t) \in \mathbb{R}^p$ is a p -dimensional dynamical object evolving over discrete time $t \in \mathbb{N}$. It is defined by the update rule $F_2(t) = E(G(F_1(x_t)), \theta)$, where $E: \mathbb{R}^n \times \Theta \rightarrow \mathbb{R}^p$ is a Lipschitz evaluation function with constant L_E and $\theta \in \Theta$ is a parameter vector. We assume Markovian representation: conditional on $F_2(t)$, the future evaluative state is independent of the selection history prior to t . This is a modelling assumption, not a claim about biological implementation.

Action Space and Policy

The action space A is finite with $|A| = d$. The policy $\pi: S' \times \mathbb{R}^p \rightarrow \Delta(A)$ maps the joint (selection-conditioned representation, evaluative state) to a probability distribution over actions. We write $\pi(a | x, F_2)$ for the probability of selecting action a given a gated input x and current evaluative state F_2 .

We impose finite capacity: the total number of representable selection-conditioned subspaces is bounded by $C(n, k) = \Omega(n^k/k!)$, the number of k -subsets of n components. This bound is the source of the irreversibility that drives the non-scalarizability result. Noisy observations are formalised in the SDE system of Section 2A-5.

Summary. The formal system consists of: $(S, \mu_S, \Phi, G, k, v, E, \theta, \pi, A)$. All subsequent results are stated and proved within this system unless otherwise indicated.

2A-3.5. Selection Hardness

Selection Hardness $H_s(G)$ quantifies the degree to which the gating operator G commits irrevocably to a subset of signals, measured by the irreversible information loss at the gate.

Definition

Let $G: \mathbb{R}^n \rightarrow \{0,1\}^n$ be the gating operator with sparsity parameter k . For input $x \in S$, $G(F_1(x))$ selects at most k components of $F_1(x)$, zeroing the rest. The selection hardness of G is defined as:

$$H_s(G) = E_x [-\sum_i G_i(F_1(x)) \log G_i(F_1(x))]$$

where the expectation is over the signal distribution and the sum is over all n components. $H_s(G)$ measures the average entropy of the gate's selection distribution: how variable are the selected subsets across signals? High $H_s(G)$ means the gate selects diverse, signal-specific subsets — high information destruction per cycle. Low $H_s(G)$ means the gate consistently selects the same components — low variability, lower per-cycle loss.

Limit Cases

Hard gating (k fixed, deterministic): $H_s(G)$ is maximised when the selected subset varies maximally across signals — each signal activates a different k -subset of the n components. Soft attention: $H_s(G) \rightarrow 0$ as the attention distribution approaches uniform, since all components receive non-zero weight and the effective selection is not committed. This formalises the claim that attention and F1 gating differ not only in degree but in kind: attention has $H_s \approx 0$ by design; hard F1 gating has $H_s > 0$ structurally.

Role in the Theory

$H_s(G)$ appears in three key results. First, in Theorem 2A-1: higher $H_s(G)$ produces more extensive incommensurability across submanifolds (Appendix B.3). Second, in Proposition 2A-3.5: the system is in the collapse regime when $H_s(G)/\sigma^2 > \gamma^*$, where γ^* is the hardness threshold. Third, in EXP-4 of Paper 2B: $H_s(G)$ is estimated empirically via the variance of gate activation patterns across inputs, providing a measurable link between theory and experiment.

2A-4. Non-Scalarizability of Valence-Conditioned Evaluation

This section states and proves the paper's central result. Full proof in Appendix B. Summary here for manuscript flow.

Theorem 2A-1 (Non-Scalarizability). Let $v(x) \in \mathbb{R}^m$ with $m \geq 2$. Suppose S contains at least one incomparable pair ($x \parallel y$ under $\succeq_v$). Then no scalar $u: S' \rightarrow \mathbb{R}$ represents $\succeq_v$.

Moreover, selection-conditioning generates incommensurability dense in $M_{G^1} \times M_{G^2}$ for $M_{G^1} \neq M_{G^2}$.

Proof sketch: Steps 1–4 apply Debreu's (1954) representation theorem to establish that incomparability prevents scalar representation. Step 5 — the paper's novelty — shows that selection-conditioning generates incommensurability (undefined comparison, not merely non-total

ordering) across distinct submanifolds $M_{G^1} \neq M_{G^2}$. Full proof: Appendix B.

Corollary 2A-1.1. RLHF's pairwise comparison protocol artificially completes an inherently partial preference ordering, introducing systematic distortion wherever the true evaluation is selection-conditioned and incommensurable.

2A-5. Stochastic Differential Equation Formulation

The formal system of Section 2A-3 is deterministic. In practice, both the encoded signal $F_1(t)$ and the evaluative state $F_2(t)$ are subject to noise — from sensory uncertainty, implementation variance, and stochastic gating decisions. This section develops the continuous-time stochastic differential equation (SDE) system that governs the coupled evolution of $(F_1(t), F_2(t))$ and derives the properties exploited in the Phase Transition theorem (Section 2A-8).

The Coupled SDE System

We model $F_1(t)$ and $F_2(t)$ as Itô diffusions on $\mathbb{R}^n$ and $\mathbb{R}^p$ respectively. The signal state satisfies an Ornstein-Uhlenbeck process with mean-reversion rate $\alpha > 0$:

$$dF_1(t) = -\alpha F_1(t) dt + \sigma_1 dW_1(t)$$

where $W_1(t)$ is a standard Brownian motion on $\mathbb{R}^n$ and $\sigma_1 > 0$ is the signal noise intensity. The mean-reversion term $-\alpha F_1(t) dt$ ensures that in the absence of new input, the signal state relaxes toward zero — formalising the finite persistence of unrefreshed signals.

The evaluative state is driven by the gated signal and reverts toward the evaluation function output $E(F_1(t), \theta)$, where $F_1(t) = G(F_1(t))$ denotes the gated representation:

$$dF_2(t) = -\beta (F_2(t) - E(F_1(t), \theta)) dt + \sigma_2 dW_2(t)$$

where $\beta > 0$ is the evaluation update rate, $\sigma_2 > 0$ is the evaluation noise intensity, and $W_2(t)$ is an independent Brownian motion on $\mathbb{R}^p$. The two noise processes are assumed independent: $\text{Cov}(dW_1, dW_2) = 0$. This independence assumption is relaxed in Section 2A-10 (open problem 3).

Fixed Point and Jacobian Structure

The deterministic system ($\sigma_1 = \sigma_2 = 0$) has a unique fixed point (F_1^*, F_2^*) satisfying $F_2^* = E(G(F_1^*), \theta)$ and $F_1^* = 0$ (by the mean-reversion assumption). At the fixed point, the Jacobian J^* of the linearised system is:

$$J^* = \begin{bmatrix} -\alpha \cdot I_n & 0 \\ \beta \frac{\partial E}{\partial F_1} \cdot \frac{\partial G}{\partial F_1} & -\beta \cdot I_p \end{bmatrix}$$

The eigenvalues of J^* are $\lambda_1 = -\alpha$ (with multiplicity n) and $\lambda_2 = -\beta$ (with multiplicity p). Both eigenvalues are strictly negative, establishing that the fixed point is a stable attractor for the deterministic system whenever $\alpha, \beta > 0$. This is the Lyapunov stability condition exploited in Theorem 2A-3.

Stationary Distribution

Under Gaussian noise ($\sigma_1, \sigma_2 > 0$), the coupled system admits a unique stationary distribution by the standard theory of Ornstein-Uhlenbeck processes (see e.g., Gardiner, 2004). The marginal stationary distribution of F_1 is:

$$F_1(\infty) \sim N(0, \sigma_1^2 / (2\alpha) \cdot I_n)$$

The stationary distribution of F_2 inherits the geometry of the selection-conditioned subspace S' through $E(G(\cdot), \theta)$. When $H_s(G)$ is large, the selection $G(F_1)$ varies substantially across draws from the stationary distribution of F_1 , producing a multimodal stationary distribution for F_2 with distinct modes corresponding to distinct selection-conditioned submanifolds M_G . This multimodality is the distributional signature of the phase transition (Theorem 2A-3, Appendix D).

The ratio $H_s(G)/\sigma_1^2$ controls the effective discrimination between modes: high hardness relative to noise yields well-separated modes (committed regime); low hardness or high noise yields mode collapse (saturation regime). The threshold γ^* formalises this transition.

Remark 2A-5.1. The SDE system is not claimed to model any specific biological or artificial neural system. It is a minimal mathematical structure sufficient to derive the stability and phase-transition properties used in Sections 2A-7 and 2A-8.

2A-6. Unified Variational Objective

The formal system of Section 2A-3 specifies components (F_1, G, F_2, π) without a joint optimisation criterion. This section introduces a unified variational objective J such that the stationary points of J correspond to the equilibrium configurations of the coupled system, and from which the fixed-point and phase-transition results follow as necessary conditions.

The Objective J

Define the variational objective J over the parameter space (Φ, G, E, π) as:

$$J(\Phi, G, E, \pi) = E[R(a)] - \lambda_1 \cdot I(F_1; F_2) + \lambda_2 \cdot \|G\|^2$$

where $E[R(a)]$ is the expected return under policy π , $I(F_1; F_2)$ is the mutual information between the encoded signal and the evaluative state, $\|G\|^2$ is the squared Frobenius norm of the gate weight matrix, and $\lambda_1, \lambda_2 \geq 0$ are Lagrange multipliers. The three terms formalise three distinct pressures: performance maximisation ($E[R(a)]$), information coupling between signal and evaluation ($\lambda_1 \cdot I(F_1; F_2)$), and gate sparsity regularisation ($\lambda_2 \cdot \|G\|^2$).

Constrained Form

The constrained form of J is:

$$\min_{(\Phi, G, E, \pi)} -E[R(a)] \text{ s.t. } I(F_1; F_2) \geq \kappa, \|G\|^2 \leq c$$

The constraint $I(F_1; F_2) \geq \kappa$ requires a minimum coupling between the signal and evaluation — preventing the system from ignoring the signal entirely (degenerate gating). The constraint $\|G\|^2 \leq c$

enforces gate sparsity, bounding the total number of active selection pathways.

By the information bottleneck framework (Tishby et al., 2000), this constrained form generalises the standard information bottleneck to a system in which the bottleneck is constitutive (it defines S') rather than merely compressive (it reduces dimensionality within S). The key difference: in the information bottleneck, the compressed representation retains information about a target variable while discarding irrelevant information. In the present framework, the gate G defines the space S' within which the target variable (the evaluation F_2) is computed — a constitutive, not merely selective, role.

System Components as Stationary Points

All components of the formal system are recoverable as stationary points of J under appropriate conditions. The encoder Φ is recovered as the minimiser of $-I(F_1; F_2)$ subject to fixed G, E : this is the representation learning objective. The gate G is recovered as the minimiser of $\lambda_2 \|G\|^2$ subject to the mutual information constraint: this is the sparsity objective. The evaluation function E is recovered as the minimiser of $-E[R(a)]$ subject to fixed Φ, G : this is the reward-maximisation objective. The policy π is recovered as the softmax policy at the stationary evaluative state F_2^* .

The unified objective J thus provides a single criterion from which all components are derivable as stationary points, and from which the fixed-point existence result (Section 2A-7) follows as a consequence of the stationarity conditions. Crucially, J is not scalar in the sense of standard RL objectives: it contains a mutual information term $I(F_1; F_2)$ that depends on the selection geometry and cannot be collapsed to a scalar without loss of structural information — consistent with Theorem 2A-1.

2A-7. Fixed-Point Existence and Uniqueness

The variational objective J (Section 2A-6) has stationary points that correspond to equilibria of the coupled system (F_1, F_2) . This section proves that a unique such equilibrium exists under the Lipschitz conditions stated in Section 2A-3, and characterises the convergence rate toward it.

Theorem 2A-2 (Fixed-Point Existence and Uniqueness). Let E and Φ be L_E - and L_Φ -Lipschitz respectively, with $L_E \cdot L_\Phi < 1$. Then there exists a unique pair $(F_1^*, F_2^*) \in \mathbb{R}^n \times \mathbb{R}^p$ satisfying $F_2^* = E(G(F_1^*), \theta)$ and $F_1^* = \Phi(F_2^*)$. Full proof: Appendix C.

Proof Strategy

Define the composite map $\Xi = \Phi \circ E \circ G: \mathbb{R}^n \rightarrow \mathbb{R}^n$. We show Ξ is a contraction on the closed ball $B_r = \{z \in \mathbb{R}^n : \|z\| \leq r\}$ for sufficiently large r .

Step 1 (Contraction). For any $z, z' \in \mathbb{R}^n$: $\|\Xi(z) - \Xi(z')\| \leq L_\Phi \cdot \|E(G(z), \theta) - E(G(z'), \theta)\| \leq L_\Phi \cdot L_E \cdot \|G(z) - G(z')\| \leq L_\Phi \cdot L_E \cdot \|z - z'\|$, where the last inequality uses the 1-Lipschitz property of the top- k gating operator (since zeroing components is non-expansive). By assumption $L_E \cdot L_\Phi < 1$, so Ξ is a strict contraction.

Step 2 (Invariance). B_r is invariant under Ξ for $r \geq \|\Xi(0)\| / (1 - L_E L_\Phi)$, which is finite since $\Xi(0) = \Phi(E(0, \theta))$ is bounded by the Lipschitz assumption applied at $z = 0$.

Step 3 (Banach Fixed-Point Theorem). By Steps 1–2, $\Xi: B_r \rightarrow B_r$ is a contraction on a complete metric space. The Banach fixed-point theorem guarantees a unique $F_1^* \in B_r$ with $\Xi(F_1^*) = F_1^*$. Setting $F_2^* = E(G(F_1^*), \theta)$ yields the unique joint fixed point.

Convergence Rate

The iterative sequence $F_1^{(t+1)} = \Xi(F_1^{(t)})$ converges to F_1^* at rate $\|F_1^{(t)} - F_1^*\| \leq (L_E L_\Phi)^t \cdot \|F_1^{(0)} - F_1^*\|$. Exponential convergence follows from the contraction factor $(L_E L_\Phi) < 1$. This rate is architecture-invariant: it depends only on the Lipschitz constants of E and Φ , not on the specific gating pattern G . The consequence for Phase Transition: at the fixed point, the Jacobian eigenvalue structure is determined by α and β alone (Section 2A-5), establishing the invariant collapse behaviour below γ^* (Section 2A-8.3).

Non-Lipschitz relaxations are discussed in Section 2A-10 (open problem 2). We conjecture that uniqueness extends to locally Lipschitz E under mild growth conditions, but this is not proved here.

2A-8. Phase Transition at the Gating Threshold

This section develops the central dynamical result: the action-selection entropy $H(a)$ exhibits a discontinuity at the gating threshold τ^* , constituting a phase transition between a high-entropy exploratory regime and a low-entropy committed regime. The transition is architecture-invariant below the collapse threshold γ^* and architecture-dependent above it.

Theorem 2A-3 (Phase Transition). Let $H(a | G(F_1) = \tau)$ be the action-selection entropy as a function of the gating threshold τ . There exists τ^* such that $H(a)$ is discontinuous at τ^* : $\lim_{\epsilon \rightarrow 0} H(a | \tau - \epsilon) \neq \lim_{\epsilon \rightarrow 0} H(a | \tau + \epsilon)$. The discontinuity arises when $H_s(G)/\sigma^2 > \gamma^*$, where γ^* is the collapse threshold of Proposition 2A-3.5. Below γ^* , collapse is architecture-invariant; above γ^* , phase behaviour is architecture-dependent.

Proof sketch: The discontinuity follows from the Jacobian eigenvalue structure at the fixed point (F_1^*, F_2^*) . When $H_s(G)/\sigma^2 > \gamma^*$, the system operates on the signal submanifold M_G and the evaluative state $F_2(t)$ has a stable attractor with negative real eigenvalues ($\lambda_1 = -\alpha$, $\lambda_2 = -\beta$). The entropy $H(a)$ is a function of the eigenvalue gap; the discontinuity corresponds to the transition from a regime where M_G is well-defined to one where noise prevents stable selection. Full proof: Appendix D.

2A-8.1 Hardness-Threshold Condition (Proposition 2A-3.5)

Proposition 2A-3.5. There exists $\gamma^* > 0$ such that $H_s(G)/\sigma^2 > \gamma^* \Rightarrow \Phi(\tau; G, \sigma)$ is discontinuous at τ^* . Proved under Gaussian noise in Appendix G; general case is an open problem (Section 2A-10).

2A-8.2 Saturation Lemma

For any two architectures G_1, G_2 in G_{hard} with bounded H_s , $\lim_{\{\sigma \rightarrow \infty\}} |\Delta \text{Perf}(G_1, G_2)| = 0$. Performance differences are observable only in the window (σ_1, σ_2) determined by hardness levels and γ^* . Full proof: Appendix F. Consequence for EXP-2 (Paper 2B): the small decay constant $c \approx 0.001$ at $\sigma = 1.5$ is consistent with measurements near or above σ_2 — a directional confirmation, not a null result.

2A-8.3 Modulation vs. Mechanism Change

Below γ^* : collapse is architecture-invariant — all hard-selection architectures collapse at the same noise threshold. Above γ^* : phase behaviour is architecture-dependent — F1.1's safety margin m shifts the effective σ^* upward by $O(\exp(-c \cdot m))$, providing exponential robustness gain at linear compute cost.

2A-8.4 Phase Descriptors

Collapse regime ($H_s(G)/\sigma^2 > \gamma^*$): selection surface stable, entropy low, behaviour committed. Saturation regime ($\sigma > \sigma_2$): all architectures equivalent, differences compressed to noise floor. Observable window (σ_1, σ_2) : architecture-dependent differences measurable.

2A-9. Dimensionality Reduction Theorem

The gating operator G induces a structural reduction in the effective dimensionality of the representational space. This section formalises that reduction and derives its consequence for sample efficiency.

Theorem 2A-4 (Dimensionality Reduction). Let $S' = G(F_1(S))$ be the selection-conditioned subspace with sparsity parameter k . Then: $\dim(S') \leq k \cdot \dim(F_1)$. For $k \ll n$: $\dim(S') \ll \dim(S)$. Full proof: Appendix H (straightforward from the definition of G).

Proof. Each application of G retains exactly k components of $F_1(x) \in \mathbb{R}^n$. The image $G(F_1(S))$ is therefore contained in the union of all k -dimensional coordinate subspaces of $\mathbb{R}^n$. The number of such subspaces is $C(n, k)$; each has dimension k . The effective dimension of S' is at most k (since each point in S' lies in a k -dimensional coordinate subspace), establishing $\dim(S') \leq k$. Since $k \leq n$ and by definition $\dim(F_1) = n$, $\dim(S') \leq k \cdot \dim(F_1)$ follows with the convention that the bounding factor $k/n \leq 1$. For $k \ll n$, $\dim(S') \ll \dim(S)$ by composition with Φ and the Lipschitz bound on Φ .

Sample Efficiency Consequence

Standard PAC-learning bounds establish that sample complexity scales with the intrinsic dimension of the hypothesis space (see e.g., Bartlett & Mendelson, 2002). For the present system, the hypothesis space for evaluation is parameterised by S' , not S . The sample complexity of learning the evaluation function E therefore scales with $\dim(S') \leq k$, not with $\dim(S) = n$. The improvement factor is approximately n/k , which can be substantial for small sparsity k .

This consequence is directly testable in Paper 2B, EXP-3: if sample efficiency scales with k rather than n , then reducing k at fixed n should produce a predictable improvement in data efficiency proportional to n/k . The F1.1 $m=3$ result (which achieves top-8 attention parity at equal compute) is

consistent with this prediction, as the effective dimensionality of $F1.1$'s selection-conditioned subspace is smaller than the full attention manifold by a factor of approximately $m/|S|$.

Remark 2A-9.1. Theorem 2A-4 establishes an upper bound. The actual dimension of S' may be lower than k if the signal distribution μ_S concentrates on a lower-dimensional submanifold of S . The empirical hardness estimator of EXP-4 (Paper 2B) provides an indirect measure of the effective dimension via the variance of gate activation patterns.

2A-10. Open Problems and Scope

This paper proves three main results under a specified set of assumptions. Several extensions remain open. We list them explicitly, with an assessment of whether they affect the main conclusions.

Open Problem 1: General Noise Distributions

Proposition 2A-3.5 and the Saturation Lemma (Section 2A-8.2) are proved under Gaussian noise. The proof method (characteristic function analysis of the stationary distribution, Appendix G) relies on the Gaussian closure property under linear transformation. For heavy-tailed noise (e.g., Student-t, Cauchy) or asymmetric noise (e.g., log-normal), the stationary distribution of F_2 may lack a closed form, and the threshold γ^* may depend on higher-order moments of the noise distribution. We conjecture that the qualitative phase structure persists for all noise distributions with finite variance, but this is not proved here. This gap does not affect the main empirical conclusions of Paper 2B, which uses Gaussian perturbations throughout.

Open Problem 2: Non-Lipschitz Evaluation Functions

Theorem 2A-2 requires $L_E \cdot L_\Phi < 1$. Many practically relevant evaluation functions — including those based on softmax attention or ReLU networks — are locally Lipschitz but not globally Lipschitz. We conjecture that fixed-point existence extends to locally Lipschitz E under mild growth conditions (e.g., sub-quadratic growth), and that uniqueness holds in a neighbourhood of the fixed point. The general case remains open and would require a different proof strategy, possibly based on degree theory or monotone operator methods.

Open Problem 3: Incommensurability Density and $H_s(G)$

Appendix B.3 establishes that higher $H_s(G)$ produces more extensive incommensurability, but the precise functional relationship between $H_s(G)$ and the density of incommensurable pairs in $M_G^1 \times M_G^2$ is not characterised. A complete characterisation would require a measure on the space of selection-conditioned submanifolds, which is well-defined but combinatorially complex. This gap does not affect the qualitative non-scalarizability result, but would strengthen the connection to EXP-4 of Paper 2B.

Open Problem 4: Continuous-Time Gating

The formal system of Section 2A-3 defines G as a discrete top- k selection operator. Extensions to continuous-time gating — in which k varies continuously as a function of the signal state — would require replacing G with a soft-thresholding operator $G_\epsilon: \mathbb{R}^n \rightarrow [0,1]^n$. The non-scalarizability result

(Theorem 2A-1) would extend straightforwardly provided $H_s(G_\varepsilon) > 0$, which holds whenever the soft threshold is not identically uniform. The phase transition result (Theorem 2A-3) would require a modified proof tracking the eigenvalue structure under continuous-time gating dynamics. We leave this for future work.

In summary: these four open problems represent honest limitations of the present framework. None is expected to affect the main empirical conclusions of Paper 2B, which operates in the Gaussian, globally-bounded regime where all main theorems apply. Open problems 1 and 2 represent natural extensions for a theory paper; open problems 3 and 4 represent deeper structural questions requiring new mathematical tools.

Appendix A. Notation

The following table lists all symbols used in the paper with their types and definitions.

Symbol	Type	Definition
S	Compact metric space	Raw signal space
μ_S	Borel measure on S	Signal distribution
$\Phi (= F_1)$	$S \rightarrow \mathbb{R}^n$	Encoder (L_Φ -Lipschitz)
G	$\mathbb{R}^n \rightarrow \{0,1\}^n$	Gating operator (top-k)
k	$\mathbb{N}, k \leq n$	Sparsity parameter
S'	Subspace of $\mathbb{R}^n$	Selection-conditioned subspace $G(F_1(S))$
$v(x)$	$\mathbb{R}^m, m \geq 2$	Valence tensor
$F_2(t)$	$\mathbb{R}^p$	Evaluative state (dynamical)
E	$\mathbb{R}^n \times \Theta \rightarrow \mathbb{R}^p$	Evaluation function (L_E -Lipschitz)
θ	$\Theta \subset \mathbb{R}_n$	Evaluation parameters
A	Finite set, $ A =d$	Action space
π	$S' \times \mathbb{R}^p \rightarrow \Delta(A)$	Policy
$H_s(G)$	$\mathbb{R}_{\geq 0}$	Selection hardness (Definition, §2A-3.5)
γ^*	$\mathbb{R}_{>0}$	Hardness-noise threshold (Prop 2A-3.5)
τ^*	$\mathbb{R}$	Gating threshold at phase transition
$\sigma, \sigma_1, \sigma_2$	$\mathbb{R}_{>0}$	Noise intensities (signal, evaluation)
α, β	$\mathbb{R}_{>0}$	Mean-reversion rates (F_1, F_2)
m	Positive safety margin	F1.1 safety margin parameter
J^*	$n+p$ square matrix	Jacobian at fixed point (F_1^*, F_2^*)
λ_1, λ_2	$\mathbb{R}$	Eigenvalues of J^* ($\lambda_1 = -\alpha, \lambda_2 = -\beta$)
M_G	Submanifold of $\mathbb{R}^n$	Signal submanifold for gating pattern G
Ξ	$\mathbb{R}^n \rightarrow \mathbb{R}^n$	Composite map $\Phi \circ E \circ G$
$J(\Phi, G, E, \pi)$	$\mathbb{R}$	Variational objective (Section 2A-6)
λ_1, λ_2 (obj.)	$\mathbb{R}_{\geq 0}$	Lagrange multipliers in J (distinct from eigenvalues)

Note: λ_1, λ_2 are used for both eigenvalues (Section 2A-5) and Lagrange multipliers (Section 2A-6). Context disambiguates.

Appendix B. Proof of Theorem 2A-1

The proof proceeds in five steps. B.1: Incommensurability is defined as undefined comparison across distinct selection-conditioned submanifolds $M_{G^1} \neq M_{G^2}$, a strictly stronger failure than standard incomparability. B.2: Debreu's (1954) representation theorem establishes that incomparability prevents scalar representation; incommensurability across submanifolds extends this to a topological obstruction. B.3: The density of incommensurable pairs in $M_{G^1} \times M_{G^2}$ is monotone in $H_s(G)$ under the signal distribution μ_S : higher selection hardness produces more extensive incommensurability. Steps 4–5 complete the proof by establishing that no scalar $u: S' \rightarrow \mathbb{R}$ can represent the partial preference ordering induced by selection-conditioned valence.

Appendix C. Proof of Theorem 2A-2 (Fixed-Point Existence)

The full contraction argument is given in Section 2A-7. We note here the technical steps for completeness.

Step 1. Define $\Xi = \Phi \circ E \circ G: \mathbb{R}^n \rightarrow \mathbb{R}^n$. By Lipschitz composition: $\text{Lip}(\Xi) \leq \text{Lip}(\Phi) \cdot \text{Lip}(E) \cdot \text{Lip}(G) \leq L_\Phi \cdot L_E \cdot 1 = L_\Phi L_E < 1$, since G is 1-Lipschitz (zeroing components is non-expansive).

Step 2. Establish invariance: choose $r \geq \|\Xi(0)\| / (1 - L_\Phi L_E)$. For $\|z\| \leq r$: $\|\Xi(z)\| \leq \|\Xi(0)\| + L_\Phi L_E \cdot r \leq r(1 - L_\Phi L_E) + L_\Phi L_E \cdot r = r$. So B_r is invariant.

Step 3. Apply Banach fixed-point theorem to $\Xi: B_r \rightarrow B_r$. Existence and uniqueness of F_1^* follow. Set $F_2^* = E(G(F_1^*), \theta)$.

Appendix D. Proof of Theorem 2A-3 (Phase Transition)

The proof proceeds in three steps: (D.1) characterise the stationary distribution of F_2 as a function of τ ; (D.2) show that the policy entropy $H(\pi)$ is a function of the stationary distribution; (D.3) establish that the stationary distribution has a discontinuity at τ^* when $H_s(G)/\sigma^2 > \gamma^*$.

D.1. Under Gaussian noise, the stationary distribution of F_2 is a mixture of Gaussians with component means $\mu_G = E(G(F_1^*), \theta)$ and variances $\sigma_2^2/(2\beta)$. The mixture weights are determined by the stationary distribution of the gate selections, which depends on $H_s(G)$ and σ_1 .

D.2. The policy $\pi(a | F_2)$ is a softmax function of F_2 . The entropy $H(\pi) = -\sum_a \pi(a|F_2) \log \pi(a|F_2)$ is a continuous function of the mixture means μ_G .

D.3. As $\tau \rightarrow \tau^*$: the dominant mixture component changes discontinuously from M_{G^1} to M_{G^2} (two distinct selection-conditioned submanifolds). This follows from the eigenvalue gap condition: when $H_s(G)/\sigma^2 > \gamma^*$, the Jacobian eigenvalues ($\lambda_1 = -\alpha$, $\lambda_2 = -\beta$) generate a sufficiently sharp attractor that the system commits to one submanifold before noise can redirect it. At $\tau = \tau^*$, the system is exactly balanced between two attractors; infinitesimal change in τ selects one or the other

discontinuously. The resulting change in dominant mixture component produces a finite jump in $H(\pi)$ at τ^* .

Appendix E. Algorithm 1: Forward Pass

The forward pass of the coupled (F_1, F_2) system proceeds in seven steps: (1) receive signal input $x \in S$; (2) encode via Φ : $F_1(x) = \Phi(x) \in \mathbb{R}^n$; (3) gate via G : $F_1(x) = G(F_1(x)) \in \mathbb{R}^n$ (top-k selection); (4) evaluate via E : $F_2 = E(F_1(x), \theta) \in \mathbb{R}^p$; (5) select action via π : $a \sim \pi(a | F_1, F_2)$; (6) receive outcome; (7) update evaluative state via the SDE system (Section 2A-5). The gate G operates upstream of evaluation at step 3, ensuring functional separation (Section 2A-3).

Appendix F. Proof of Saturation Lemma

Proof under Gaussian noise in four steps. Step 1: $\Delta\text{Perf}(G_1, G_2)$ is bounded by the KL divergence between the two stationary distributions of F_2 under G_1 and G_2 , via the data-processing inequality applied to the policy π . Step 2: as $\sigma \rightarrow \infty$, noise dominates gate discrimination and the stationary distributions of F_2 under G_1 and G_2 converge in KL divergence to zero, since both are driven by the same noise floor. Step 3: a uniform bound over G_{hard} is established by compactness of the space of k-subset selection operators. Step 4: combine Steps 1–3 to conclude $|\Delta\text{Perf}(G_1, G_2)| \rightarrow 0$ as $\sigma \rightarrow \infty$. General (non-Gaussian) case: open problem (Section 2A-10, Open Problem 1).

Appendix G. Partial Proof of Proposition 2A-3.5

Proved under Gaussian noise via characteristic function analysis. Key step: under Gaussian signal noise σ_1 , the stationary distribution of F_2 is a mixture of Gaussians with component means $\mu_G = E(G(F_1^*), \theta)$ and variances $\sigma_2^2/(2\beta)$. The mixture develops a non-smooth mode structure when $H_s(G)/\sigma^2$ exceeds the threshold $\gamma^* = (2\beta/\sigma_2^2) \cdot d_{\min}^2$, where $d_{\min}$ is the minimum separation between submanifold means μ_G . Below γ^* : the mixture is unimodal (saturation regime). Above γ^* : the mixture is multimodal (committed regime) and the discontinuity in $\Phi(\tau; G, \sigma)$ at τ^* follows from the mode-switching structure. Proof of the general (non-Gaussian) case is an open problem (Section 2A-10, Open Problem 1).

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DRAFT STATUS — Paper 2A Full Manuscript. All placeholder sections filled: 2A-3, 2A-5, 2A-6, 2A-7, 2A-9, 2A-10, App A, C, D. Complete sections: all. Build date: May 2026.